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Inventor(s)	Pivetta; Martin J. C.

Safer password manager, trusted services, and anti-phishing process

Abstract

Preventing anomalous connections includes detecting by a programmable device an attempt by a first device to connect to a second device, detecting a first connection anomaly responsive to characteristics of the first device and characteristics of one or more other devices, and prohibiting a connection between the first device and the second device responsive to detecting the first connection anomaly.

Inventors: **Pivetta; Martin J. C. (Luxemborg, BE)**

Applicant: **McAfee, LLC** (Santa Clara, CA)

Family ID: **59086722**

Assignee: **McAfee, LLC (San Jose, CA)**

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Primary Examiner: Homayounmehr; Farid

Assistant Examiner: Debnath; Suman

Attorney, Agent or Firm: Hanley, Flight & Zimmerman, LLC

Background/Summary

RELATED APPLICATION (1) This patent arises from a continuation of U.S. patent application Ser. No. 14/757,599, (now U.S. Pat. No. 10,523,702) which was filed on Dec. 23, 2015. U.S. patent application Ser. No. 14/757,599 is hereby incorporated herein by reference in its entirety. Priority to U.S. patent application Ser. No. 14/757,599 is hereby claimed.

TECHNICAL FIELD
(1) Embodiments described herein generally relate to data collection, and more specifically to sensor data collection, protection, and value extraction.
BACKGROUND ART
(2) With the rise of the Internet users can access a multitude of devices from their client device. These remote devices can provide valuable services, such as banking, retail, email, social networking, and the like. However, utilizing these types of services comes with risk. As an example, malicious entities may attempt to acquire sensitive information such as credit card numbers, user names, passwords, configuration details, user data, and the like by mimicking a legitimate service.
(3) Current solutions consider a URL, or basic service characteristic as a trustworthy reference point for defining the trustworthiness of the remote connection. However, security holes still exist. Man-in-the-middle attack protection tools leverage local information to detect a compromise or depend on the end-user to spot the issue or know what to do in a situation. Many systems also depend on the first access to a remote destination to determine whether future attempts to access the remote destination should be trusted. Some password managers may merely check a URL of a website prior to providing password information, without further analysis of the server behind the URL.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a diagram illustrating a network of programmable devices according to one or more embodiments.
(2) FIG. **2** is a system diagram illustrating a network diagram for a safer Password Manager,

Trusted Services, and Anti-Phishing process, according to one or more embodiments.

(3) FIG. **3** is a flow diagram illustrating a method for providing a safer Password Manager, Trusted Services, and Anti-Phishing process, according to one or more embodiments.

(4) FIG. **4** is a diagram illustrating a computing device for use with techniques described herein according to one embodiment.

(5) FIG. **5** is a block diagram illustrating a computing device for use with techniques described herein according to another embodiment.

DESCRIPTION

(6) In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the invention. It will be apparent, however, to one skilled in the art that the invention may be practiced without these specific details. In other instances, structure and devices are shown in block diagram form in order to avoid obscuring the invention. References to numbers without subscripts or suffixes are understood to reference all instance of subscripts and suffixes corresponding to the referenced number. Moreover, the language used in this disclosure has been principally selected for readability and instructional purposes, and may not have been selected to delineate or circumscribe the inventive subject matter, resort to the claims being necessary to determine such inventive subject matter. Reference in the specification to “one embodiment” or to “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least one embodiment of the invention, and multiple references to “one embodiment” or “an embodiment” should not be understood as necessarily all referring to the same embodiment.

(7) As used herein, the term “computer system” can refer to a single computer or a plurality of computers working together to perform the function described as being performed on or by a computer system.

(8) As used herein, the term “medium” refers to a single physical medium or a plurality of media that together store what is described as being stored on the medium.

(9) As used herein, the term “network device” can refer to any computer system that is capable of communicating with another computer system across any type of network.

(10) In one or more embodiments, a technique is provided for a safer Password Manager, Trusted Services, and Anti-Phishing process. In one or more embodiments, an intermediary device or system may detect that a first device is attempting to connect to a second device. In one or more embodiments, the first device may be a client device whereas the second device may be a server associated with a web service or website. The intermediary device may identify attempted connection characteristics between the first and second device. For example, the intermediary device may obtain information such as a location of the server the first device is attempting to connect with, an IP address for the second device, security information between the first device and second device, and the like. The intermediary device may also obtain historic connection characteristics between the first and second device. In one or more embodiments, the intermediary device may also attempt to connect to the second device itself to obtain its own connection characteristics with the second device. In addition, in one or more embodiments, the intermediary device may also obtain historic characteristics between other devices and the second device. For example, the intermediary device may obtain connection characteristics arising from connections between other client devices and the web server within a predetermined time period. The intermediary device may determine a trustworthiness of the second device based on the connection characteristics. In one or more embodiments, the intermediary device may look for anomalies in the attempted connection between the first device and the second device, as compared to the other obtained connection characteristics. The intermediary device may either allow or prohibit the connection between the first device and the second device based on the determination.

(11) Referring to the figures, FIG. **1** an example infrastructure **100** in which embodiments may be implemented is illustrated schematically. Infrastructure **100** contains computer networks **102**.

Computer networks **102** may include many different types of computer networks available today, such as the Internet, a corporate network, or a Local Area Network (LAN). Each of these networks can contain wired or wireless programmable devices and operate using any number of network protocols (e.g., TCP/IP). Networks **102** may be connected to gateways and routers (represented by **108**), end user computers **106**, and computer servers **104**. Infrastructure **100** also includes cellular network **103** for use with mobile communication devices. Mobile cellular networks support mobile phones and many other types of mobile devices. Mobile devices in the infrastructure **100** are illustrated as mobile phones **110**, laptops **112**, and tablets **114**. A mobile device such as mobile phone **110** may interact with one or more mobile provider networks as the mobile device moves, typically interacting with a plurality of mobile network towers **120**, **130**, and **140** for connecting to the cellular network **103**. Each of the networks **102** may contain a number of other devices typically referred to as Internet of Things (microcontrollers, embedded systems, industrial control computing modules, etc.). Although referred to as a cellular network in FIG. **1**, a mobile device may interact with towers of more than one provider network, as well as with multiple non-cellular devices such as wireless access points and routers **108**. In addition, the mobile devices **110**, **112**, and **114** may interact with non-mobile devices such as computers **104** and **106** for desired services. The functionality of the gateway device **108** may be implemented in any device or combination of devices illustrated in FIG. **1**; however, most commonly is implemented in a firewall or intrusion protection system in a gateway or router.

(12) FIG. **2** is a diagram illustrating a system for safer password managers and anti-phishing process, according to one or more embodiments. FIG. **2** includes several devices connected across network **200**. As depicted, network **200** may connect such computer systems as a security system **205**, client device **210**, additional devices **215**A-**215**N, web server **220**, and network storage **225**. In one or more embodiments, the functionality of the various components may be differently distributed than the particular depiction of FIG. **2**. Network **200** may be any type of computer network, such as a LAN or a corporate network, including a plurality of interconnected networks. For example, Network **200** may include a subset of the devices included in larger network **102** or **103**.

(13) In one or more embodiments, the client device **210** may be a computer device with numerous components. Further, in one or more embodiments, the client device **210**, as depicted, may be a more constricted device, and have only bare bones components, such as a processor **235** and memory **230**. Processor **235** and memory **230** may be operatively connected, for example, over a bus. In one or more embodiments, memory **230** may include software executable by the processor **235**, such as a web browser **240**. In one or more embodiments, the web browser **240** may allow client device **210** to connect to other devices across network **200**.

(14) In one or more embodiments, client device **210**, or additional devices **215**A-**215**N could utilize web services across network **200**. In one or more embodiments, the various devices, such as client device **210** may connect to a web service, for example, through a web browser **240**. In one or more embodiments, the client device **210** or additional devices **215**A-**215**N may connect to web server **220** across network **200**. Although web server **220** is depicted as a single server, web server **220** could include any number of network devices utilized to provide a web service.

(15) In one or more embodiments, the security system **205** acts as an intermediary device between at least some of client device **210** and additional devices **215**A-**215**N and web server **220**. Although security system **205** is depicted as a single computer system, the various components and functionalities of security system **205** could be distributed among multiple computer systems, servers, and the like. As depicted, security system **205** may include a memory **245**, processor **255**, and storage **250**. Memory **245**, processor **255**, and storage **250** may be connected, for example, over a bus.

(16) In one or more embodiments, memory **245** includes modules such as a web browser **260** and a security module **265**. Browser **260** and security module **265** may be computer code, such as

software, executable by processor **255**. In one or more embodiments, browser **260** may allow client device **210** to connect to other devices across network **200**. For example, browser **260** may allow the security system **205** to connect to web server **220** across network **200**.

(17) In one or more embodiments, security module **265** may determine trustworthiness of a destination device on behalf of a requesting device **210**. Trustworthiness may be determined in response to detecting an attempted connection between a source device and a destination device. In one or more embodiments, the detection may additionally, or alternatively occur prior to or in conjunction with a password manager automatically entering user data into a web form. In one or more embodiments, the security module **265** may compare connection characteristics from the viewpoint of the requesting device, to connection characteristics of other devices connecting to or that have connected to the destination device. If an anomaly is detected, the destination device may be untrustworthy. For example, in one or more embodiments, security module **265** may detect that client device **210** is attempting to connect to a second device, such as web server **220**. Security system **205** may interrupt the connection to gather connection characteristics for the web server **220** to determine whether the web server **220** is trustworthy. Connection characteristics may include, for example, an IP address for the destination, a location of the destination, a certificate for the destination, security checks from the web server **220**, and the like. In one or more embodiments, the connection characteristics as viewed from the requesting device are compared to connection characteristics between other devices and the destination device, or historic connection characteristics between the requesting device and the destination device. In one or more embodiments, if an anomaly is detected between the current attempted connection characteristics and the other connection characteristics, the security module **265** may determine that the destination device is untrustworthy.

(18) In one or more embodiments, the security system **205** may itself connect or attempt to connect to the destination device, such as web server **220**, to obtain connection characteristics to compare with the attempted connection characteristics between the requesting device and the destination device. Thus, if client device **210** is attempting to connect to web server **220**, and the connection is interrupted by security system **205**, then security system **205** may connect or attempt to connect to web server **220**, for example using browser **260**, to collect additional connection characteristics regarding the web server **220**.

(19) Security module **265** may also obtain data from other devices that are connected to or have connected to the destination device. For example, security module **265** may obtain connection characteristics from connections between any of additional devices **215**A-**215**N and web server **220**. In one or more embodiments, security module **265** may consider only some connection data from other devices. For example, security module **265** may consider connection data only from other devices from within a predetermined time interval. That is, in one or more embodiments, the security module **265** may consider only recent or current connection data, or may compare connection data from a similar time of day. Security module **265** may also consider only connection data from devices from a similar location. For example, if client device **210** is within the United States, and Additional Device **215**A is also within the United States, but Additional Device **215**N is in Europe, it may be more beneficial to consider connection data only from Additional Device **215**A. Security Module **265** may also consider historic connection data between any device, including the security system **205**, client device **210**, and additional devices **215**A-**215**N and the web server **220**.

(20) In one or more embodiments, connection information be stored, for example, locally within a connection data store **270** in storage **250** for the security system **205**. In one or more embodiments, connection data may additionally or alternatively be stored remotely, such as in network storage **225**, as depicted by historic connection data **275**. Connection data may be stored in any kind of file structure, such as a database. Further, connection data may be stored for any device connecting to a destination. That is connection data **270** or historic connection data **275** may, for example, include

connection data from any of client device **210**, security system **205**, and additional devices **215**A-**215**N connecting to web server **220**. In one or more embodiments, the historic connection data relating to a particular destination may be used to determine trends regarding the destination. For example, aggregated connection data regarding web server **220** may indicate a reliability of an IP address associated with web server **220**.

(21) FIG. **3** is a flow diagram illustrating a technique for safer password managers and anti-phishing process. In one or more embodiments, the various steps of the flow diagram may occur in an intermediary device, such as security system **205** depicted in FIG. **2**. In one or more embodiments, at least some of the actions depicted in the flow diagram may be performed, for example, by security module **265** of security system **205**. Although the various steps are depicted in a particular order, and performed by particular components, in one or more embodiments, the various steps may be performed in a different order. Further, in one or more embodiments, steps may be omitted or added, or performed by different components than those depicted in FIG. **3**.

(22) The flow diagram begins at **302**, and security system **205** detects that a first device is attempting to connect to a second device. As an example, client device **210** could be attempting to connect to web server **220**. In one or more embodiments, client device **210** may indicate to security system **205** that it is attempting to connect to web server **220**. For example, browser **240** may include a security plugin that prevents immediate connection to some or all remote devices prior to alerting security system **205** and perhaps receiving permission from security system **205**.

(23) The flow diagram continues at **304**, and the security module **265** obtains connection characteristics for the second device. As described above, the connection characteristics can be obtained from a number of source devices that have connected or attempted to connect to the same second device. As depicted in the flow diagram, obtaining connection characteristics can include a number of steps, although in one or more embodiments, any combination of those shown may be utilized to obtain connection characteristics for the second device.

(24) Obtaining connection characteristics for the second device may include, at **306**, the security module **265** identifying attempted connection characteristics between the first and second device. For example, if client device **210** is attempting to connect to web server **220**, security module **265** may collect connection information from the point of view of the client device **210**. That is, security module **265** may collect information such as a location of the client device **210**, a location of the web server **220**, a certificate the client device **210** obtains from the web server **220**, a type of server of the web server **220** (such as operating system characteristics), TCP IP headers, and the like. The flow diagram continues at **308**, and the security module **265** may obtain historic connection characteristics between the first device and the second device. In one or more embodiments, the historic connection characteristics may be stored locally, such as in connection data **270**, or remotely, such as in historic connection data **275**. In one or more embodiments, historic connection characteristics between the first device and the second device may be useful for determining, for example, if this particular connection between the first and second device includes an anomaly. As an example, if the first device always connects to a web server in the United States, but now the web server appears to be in China, an anomaly may be detected.

(25) The flow diagram continues at **310**, and the security system **205** attempts a connection to the second device, such as web server **220**. In one or more embodiments, the security system **205** connects to web server **220** so that at **312**, security system **205** can obtain additional connection characteristics for the second device. That is, at **312**, the security system **205** may collect its own connection characteristics for the second device from the point of view of the security system. In one or more embodiments, the security system **205** may collect the connection characteristics in real time, or may alternatively, or additionally, rely on historic connection results from connections between the security system **205** and web server **220**. That is, in one or more embodiments, security system **205** may occasionally or periodically connect to web server **220** for purposes of gathering connection data.

(26) The flow diagram continues at **314**, and the security module **265** may additionally collect connection characteristics from connections or attempted connections between other devices and the second device. For example, security module **265** may obtain connection data from connections between additional devices **215**A-**215**N and web server **220**. The connection characteristics may be connection characteristics from current connections and obtained in real time, or may be obtained from storage of historic connection data **275**. For example, security module **265** may obtain connection characteristics from connections between additional devices **215**A-**215**N and web server **220** within connection data **270** and historic connection data **275**.
(27) The flow diagram continues at **316**, and the security system **205** determines a trustworthiness of the second device based on the obtained connection characteristics for the second device. For example, security module **265** may determine the trustworthiness of web server **220** based on any combination of connection characteristics obtained from steps **306-314**, as described above. In one or more embodiments, trustworthiness may be defined on a scale, and if the trustworthiness matches a particular threshold on the trustworthiness scale, the second device may be determined to be trustworthy. In one or more embodiments, trustworthiness may be determined by detecting anomalies in the current connection characteristics between the first and second device and the other obtained connection characteristics.
(28) The flow diagram continues at **318**, and if the second device is determined to be sufficiently trustworthy, then the flow diagram continues at **320** and the security system **205** allows the connection between the first device and the second device. However, if at **318** it is determined that the second device is not sufficiently trustworthy, then the flow diagram continues at **322** and the security system **205** prohibits the connection between the first and second device. In one or more embodiments, the security system **205** may block the connection between the first device and the second device, or the security system **205** may transmit an alert message to client device **210** to make client device **210** aware that the second device is not sufficiently trustworthy. In one or more embodiments, the security system **205** may prevent other applications, such as password managers, from transmitting data to the second device. Further, in one or more embodiments, the security system **205** may take some other action in response to determining that the second device is determined to be untrustworthy.
(29) Referring now to FIG. **4**, a block diagram illustrates a programmable device **600** that may be used within a computer device, such as security system **205**, client device **210**, or any other computer system described above in accordance with one or more embodiments. The programmable device **600** illustrated in FIG. **4** is a multiprocessor programmable device that includes a first processing element **670** and a second processing element **680**. While two processing elements **670** and **680** are shown, an embodiment of programmable device **600** may also include only one such processing element.
(30) Programmable device **600** is illustrated as a point-to-point interconnect system, in which the first processing element **670** and second processing element **680** are coupled via a point-to-point interconnect **650**. Any or all of the interconnects illustrated in FIG. **4** may be implemented as a multi-drop bus rather than point-to-point interconnects.
(31) As illustrated in FIG. **4**, each of processing elements **670** and **680** may be multicore processors, including first and second processor cores (i.e., processor cores **674***a* and **674***b* and processor cores **684***a* and **684***b*). Such cores **674***a*, **674***b*, **684***a*, **684***b* may be configured to execute instruction code in a manner similar to that discussed above in connection with FIGS. **1-3**. However, other embodiments may use processing elements that are single core processors as desired. In embodiments with multiple processing elements **670**, **680**, each processing element may be implemented with different numbers of cores as desired.
(32) Each processing element **670**, **680** may include at least one shared cache **646**. The shared cache **646***a*, **646***b* may store data (e.g., instructions) that are utilized by one or more components of the processing element, such as the cores **674***a*, **674***b* and **684***a*, **684***b*, respectively. For example,

the shared cache may locally cache data stored in a memory **632**, **634** for faster access by components of the processing elements **670**, **680**. In one or more embodiments, the shared cache **646***a*, **646***b* may include one or more mid-level caches, such as level 2 (L2), level 3 (L3), level 4 (L4), or other levels of cache, a last level cache (LLC), or combinations thereof.
(33) While FIG. **4** illustrates a programmable device with two processing elements **670**, **680** for clarity of the drawing, the scope of the present invention is not so limited and any number of processing elements may be present. Alternatively, one or more of processing elements **670**, **680** may be an element other than a processor, such as an graphics processing unit (GPU), a digital signal processing (DSP) unit, a field programmable gate array, or any other programmable processing element. Processing element **680** may be heterogeneous or asymmetric to processing element **670**. There may be a variety of differences between processing elements **670**, **680** in terms of a spectrum of metrics of merit including architectural, microarchitectural, thermal, power consumption characteristics, and the like. These differences may effectively manifest themselves as asymmetry and heterogeneity amongst processing elements **670**, **680**. In some embodiments, the various processing elements **670**, **680** may reside in the same die package.
(34) First processing element **670** may further include memory controller logic (MC) **672** and point-to-point (PP) interconnects **676** and **678**. Similarly, second processing element **680** may include a MC **682** and P-P interconnects **686** and **688**. As illustrated in FIG. **4**, MCs **672** and **682** couple processing elements **670**, **680** to respective memories, namely a memory **632** and a memory **634**, which may be portions of main memory locally attached to the respective processors. While MC logic **672** and **682** is illustrated as integrated into processing elements **670**, **680**, in some embodiments the memory controller logic may be discrete logic outside processing elements **670**, **680** rather than integrated therein.
(35) Processing element **670** and processing element **680** may be coupled to an I/O subsystem **690** via respective P-P interconnects **676** and **686** through links **652** and **654**. As illustrated in FIG. **4**, I/O subsystem **690** includes P-P interconnects **694** and **698**. Furthermore, I/O subsystem **690** includes an interface **692** to couple I/O subsystem **690** with a high performance graphics engine **638**. In one embodiment, a bus (not shown) may be used to couple graphics engine **638** to I/O subsystem **690**. Alternately, a point-to-point interconnect **639** may couple these components.
(36) In turn, I/O subsystem **690** may be coupled to a first link **616** via an interface **696**. In one embodiment, first link **616** may be a Peripheral Component Interconnect (PCI) bus, or a bus such as a PCI Express bus or another I/O interconnect bus, although the scope of the present invention is not so limited.
(37) As illustrated in FIG. **4**, various I/O devices **614**, **624** may be coupled to first link **616**, along with a bridge **618** which may couple first link **616** to a second link **620**. In one embodiment, second link **620** may be a low pin count (LPC) bus. Various devices may be coupled to second link **620** including, for example, a keyboard/mouse **612**, communication device(s) **626** (which may in turn be in communication with the computer network **603**), and a data storage unit **628** such as a disk drive or other mass storage device which may include code **630**, in one embodiment. The code **630** may include instructions for performing embodiments of one or more of the techniques described above. Further, an audio I/O **624** may be coupled to second bus **620**.
(38) Note that other embodiments are contemplated. For example, instead of the point-to-point architecture of FIG. **4**, a system may implement a multi-drop bus or another such communication topology. Although links **616** and **620** are illustrated as busses in FIG. **4**, any desired type of link may be used. Also, the elements of FIG. **4** may alternatively be partitioned using more or fewer integrated chips than illustrated in FIG. **4**.
(39) Referring now to FIG. **5**, a block diagram illustrates a programmable device **700** according to another embodiment. Certain aspects of FIG. **4** have been omitted from FIG. **5** in order to avoid obscuring other aspects of FIG. **5**.
(40) FIG. **5** illustrates that processing elements **770**, **780** may include integrated memory and I/O

control logic ("CL") **772** and **782**, respectively. In some embodiments, the **772**, **782** may include memory control logic (MC) such as that described above in connection with FIG. **4**. In addition, CL **772**, **782** may also include I/O control logic. FIG. **5** illustrates that not only may the memories **732**, **734** be coupled to the **772**, **782**, but also that I/O devices **744** may also be coupled to the control logic **772**, **782**. Legacy I/O devices **715** may be coupled to the I/O subsystem **790** by interface **796**. Each processing element **770**, **780** may include multiple processor cores, illustrated in FIG. **5** as processor cores **774**A, **774**B, **784**A, and **784**B. As illustrated in FIG. **5**, I/O subsystem **790** includes P-P interconnects **794** and **798** that connect to P-P interconnects **776** and **786** of the processing elements **770** and **780** with links **752** and **754**. Processing elements **770** and **780** may also be interconnected by link **750** and interconnects **778** and **788**, respectively.

(41) The programmable devices depicted in FIGS. **4** and **5** are schematic illustrations of embodiments of programmable devices which may be utilized to implement various embodiments discussed herein. Various components of the programmable devices depicted in FIGS. **4** and **5** may be combined in a system-on-a-chip (SoC) architecture.

(42) In one or more embodiments, the features described above provide safer password managers and anti-phishing processes. A security system may act as an intermediary to provide an extra layer of protection such that even if security products installed on a device identify a remote connection as trustworthy, the security system does an additional check, utilizing connection information from not only the requesting device but other devices that have connected to the remote device. Further, in one or more embodiments, the security system may prevent man in the middle attacks by leveraging aggregated connection information for a destination and comparing the aggregated connection information with the data as viewed from the requesting device.

(43) It is to be understood that the various components of the flow diagrams described above, could occur in a different order or even concurrently. It should also be understood that various embodiments of the inventions may include all or just some of the components described above. Thus, the flow diagrams are provided for better understanding of the embodiments, but the specific ordering of the components of the flow diagrams are not intended to be limiting unless otherwise described so.

(44) Program instructions may be used to cause a general-purpose or special-purpose processing system that is programmed with the instructions to perform the operations described herein. Alternatively, the operations may be performed by specific hardware components that contain hardwired logic for performing the operations, or by any combination of programmed computer components and custom hardware components. The methods described herein may be provided as a computer program product that may include a machine readable medium having stored thereon instructions that may be used to program a processing system or other electronic device to perform the methods. The term "machine readable medium" used herein shall include any medium that is capable of storing or encoding a sequence of instructions for execution by the machine and that cause the machine to perform any one of the methods described herein. The term "machine readable medium" shall accordingly include, but not be limited to, tangible, non-transitory memories such as solid-state memories, optical and magnetic disks. Furthermore, it is common in the art to speak of software, in one form or another (e.g., program, procedure, process, application, module, logic, and so on) as taking an action or causing a result. Such expressions are merely a shorthand way of stating that the execution of the software by a processing system causes the processor to perform an action or produce a result.

(45) The following examples pertain to further embodiments.

(46) Example 1 is a machine readable medium on which instructions are stored, comprising instructions that when executed cause a machine to: detect that a first device is attempting to connect to a second device; determine a level of trustworthiness of the second device based on characteristics of the first device and characteristics of additional devices connected to the second device; and in response to determining that the second device is not sufficiently trustworthy,

prevent the first device from connecting to the second device.
(47) In Example 2 the subject matter of Example 1 optionally includes further comprising instructions that when executed cause the machine to: attempt a connection between the machine and the second device, wherein the instructions that when executed cause the machine to determine a level of trustworthiness further comprise instructions that when executed cause the machine to determine a level of trustworthiness further based on characteristics of the attempted connection between the machine and the second device.
(48) In Example 3 the subject matter of any of Examples 1-2 optionally includes wherein the characteristics of the first device comprise characteristics of previous connections between the first device and the second device.
(49) In Example 4 the subject matter of any of Examples 1-2 optionally includes wherein the characteristics of the additional devices comprise characteristics of additional devices that have connected to the second device within a predetermined prior time period.
(50) In Example 5 the subject matter of any of Examples 1-2 optionally includes wherein the instructions that when executed cause the machine to determine the level of trustworthiness comprise instructions that when executed cause the machine to determine a level of trustworthiness further based on a certificate for the second device.
(51) In Example 6 the subject matter of any of Examples 1-2 optionally includes wherein the instructions that when executed cause the machine to determine the level of trustworthiness further comprise instructions that when executed determine a reliability of an IP address associated with the second device.
(52) In Example 7 the subject matter of any of Examples 1-2 optionally includes wherein the characteristics of the additional devices comprise characteristics of connections between the additional devices and the second device.
(53) In Example 8 the subject matter of Example 7 optionally includes wherein the connections between the additional devices and the second device comprise connections between the additional devices and the second device during a predetermined prior time period.
(54) Example 9 is a method of preventing anomalous connections, comprising: detecting by a programmable device an attempt by a first device to connect to a second device; detecting a first connection anomaly responsive to characteristics of the first device and characteristics of one or more other devices; and prohibiting a connection between the first device and the second device responsive to detecting the first connection anomaly.
(55) In Example 10 the subject matter of Example 9 optionally includes wherein the characteristics of the one or more other devices comprise characteristics of connections between the one or more other devices and the second device.
(56) In Example 11 the subject matter of Example 10 optionally includes wherein the connections between the one or more other devices and the second device comprise connections between the one or more other devices and the second device that were established during a predetermined prior time interval.
(57) In Example 12 the subject matter of any of Examples 9-11 optionally includes further comprising: attempting a second connection between the programmable device and the second device, wherein detecting the first connection anomaly comprises detecting the first connection anomaly responsive to characteristics of the second connection, the characteristics of the first device, and the characteristics of the one or more other devices.
(58) In Example 13 the subject matter of Example 12 optionally includes wherein detecting the first connection anomaly further comprises detecting the first connection anomaly responsive to characteristics of prior connections between the programmable device and the second device.
(59) In Example 14 the subject matter of any of Examples 9-11 optionally includes wherein the characteristics of the first device comprise characteristics of prior connections between the first device and the second device.

(60) In Example 15 the subject matter of any of Examples 9-11 optionally includes wherein detecting the first connection anomaly comprises detecting the first connection anomaly responsive to the characteristics of the first device, the characteristics of the one or more other devices; and the characteristics of the second device.
(61) In Example 16 the subject matter of Example 15 optionally includes wherein the characteristics of the second device comprise characteristics of a certificate associated with the second device.
(62) Example 17 is a system for detecting connection anomalies, comprising: one or more processors; a memory on which are stored instructions, comprising instructions that when executed cause at least some of the one or more processors to: detect that a first device is attempting to connect to a second device; determine a level of trustworthiness of the second device based on characteristics of the first device and characteristics of additional devices previously connected to the second device; and take an action regarding the attempt by the first device to connect to the second device responsive to the determined level of trustworthiness of the second device.
(63) In Example 18 the subject matter of Example 17 optionally includes wherein the instructions further comprise instructions that when executed cause at least some of the one or more processors to: attempt a connection between the system and the second device, wherein the instructions that when executed cause at least some of the one or more processors to determine a level of trustworthiness further comprise instructions that when executed cause at least some of the one or more processors to determine the level of trustworthiness further based on characteristics of the attempted connection between the system and the second device.
(64) In Example 19 the subject matter of any of Examples 17-18 optionally includes wherein the characteristics of the first device comprise characteristics of previous connections between the first device and the second device.
(65) In Example 20 the subject matter of any of Examples 17-18 optionally includes wherein the characteristics of the additional devices comprise characteristics of additional devices that have connected to the second device within a predetermined prior time period.
(66) In Example 21 the subject matter of any of Examples 17-18 optionally includes wherein the instructions that when executed cause at least some of the one or more processors to determine a level of trustworthiness comprise instructions that when executed cause at least some of the one or more processors to determine the level of trustworthiness further based on a certificate for the second device.
(67) In Example 22 the subject matter of any of Examples 17-18 optionally includes wherein the instructions that when executed cause at least some of the one or more processors to determine a level of trustworthiness further comprise instructions that when executed cause at least some of the one or more processors to determine a reliability of an IP address associated with the second device.
(68) In Example 23 the subject matter of any of Examples 17-18 optionally includes wherein the characteristics of the additional devices comprise characteristics of connections between the additional devices and the second device.
(69) In Example 24 the subject matter of Example 23 optionally includes wherein the connections between the additional devices and the second device comprise connections between the additional devices and the second device during a predetermined prior time period
(70) In Example 25 the subject matter of any of Examples 17-18 optionally includes where the instructions then when executed cause at least some of the one or more processors to take an action comprise instructions that when executed cause at least some of the one or more processors to prevent the first device from connecting to the second device.
(71) Example 26 is a method for detecting connection anomalies, comprising: detecting that a first device is attempting to connect to a second device; determining a level of trustworthiness of the second device based on characteristics of the first device and characteristics of additional devices

previously connected to the second device; and taking an action regarding the attempt by the first device to connect to the second device responsive to the determined level of trustworthiness of the second device.
(72) In Example 27 the subject matter of Example 26 optionally includes further comprising: attempting a connection between the system and the second device, wherein determining the level of trustworthiness is further based on characteristics of the attempted connection between the system and the second device.
(73) In Example 28 the subject matter of any of Examples 26-27 optionally includes wherein the characteristics of the first device comprise characteristics of previous connections between the first device and the second device.
(74) In Example 29 the subject matter of any of Examples 26-27 optionally includes wherein the characteristics of the additional devices comprise characteristics of additional devices that have connected to the second device within a predetermined prior time period.
(75) In Example 30 the subject matter of any of Examples 26-27 optionally includes wherein determining a level of trustworthiness further comprises determining the level of trustworthiness further based on a certificate for the second device.
(76) In Example 31 the subject matter of any of Examples 26-27 optionally includes wherein determining a level of trustworthiness further comprises determining a reliability of an IP address associated with the second device.
(77) In Example 32 the subject matter of any of Examples 26-27 optionally includes wherein the characteristics of the additional devices comprise characteristics of connections between the additional devices and the second device.
(78) It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments may be used in combination with each other. As another example, the above-described flow diagrams include a series of actions which may not be performed in the particular order depicted in the drawings. Rather, the various actions may occur in a different order, or even simultaneously. Many other embodiment will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should therefore should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. At least one storage device or storage disk comprising instructions to cause one or more processor circuits of a security device to at least: prevent a first communication from reaching a receiving device, the first communication from a sending device and addressed to the receiving device; initiate sending of a second communication from the security device to the receiving device; collect connection characteristics associated with the second communication between the security device and the receiving device; determine a level of trustworthiness of the receiving device based on the connection characteristics associated with the second communication between the security device and the receiving device; compare the level of trustworthiness of the receiving device to a threshold of trustworthiness; and after a determination that the level of trustworthiness of the receiving device meets the threshold, (1) notify the sending device that a third communication addressed to the receiving device is permitted and (2) permit the third communication to reach the receiving device.
2. The storage device or storage disk of claim 1, wherein at least one of the one or more processor circuits is to initiate the sending of the second communication within a predetermined interval of the first communication.
3. At least one storage device or storage disk comprising instructions to cause one or more processor circuits of a security device to at least: prevent a first communication from reaching a

receiving device, the first communication from a sending device and addressed to the receiving device; initiate sending of a second communication from the security device to the receiving device; collect connection characteristics associated with the second communication between the security device and the receiving device; determine a level of trustworthiness of the receiving device based on the connection characteristics associated with the second communication between the security device and the receiving device; compare the level of trustworthiness of the receiving device to a threshold of trustworthiness; and after a determination that the level of trustworthiness of the receiving device does not meet the threshold, disallow a third communication from the sending device from reaching the receiving device.
4. The storage device or storage disk of claim 3, wherein the instructions cause at least one of the one or more processor circuits of the security device to disallow the third communication between the receiving device and the sending device by blocking at least one communication from the sending device to the receiving device.
5. The storage device or storage disk of claim 3, wherein the instructions cause at least one of the one or more processor circuits of the security device to disallow the third communication between the receiving device and the sending device by preventing an application of the sending device from transmitting data to the receiving device.
6. The storage device or storage disk of claim 3, wherein the instructions cause at least one of the one or more processor circuits of the security device to instruct the sending device not to connect to the receiving device.
7. The storage device or storage disk of claim 3, wherein the connection characteristics are first connection characteristics, and at least one of the one or more processor circuits is to determine the level of trustworthiness of the receiving device based on the first connection characteristics associated with the second communication and second connection characteristics associated with a fourth communication including a peer device and the receiving device.
8. The storage device or storage disk of claim 3, wherein the connection characteristics are first connection characteristics, and at least one of the one or more processor circuits is to determine the level of trustworthiness based on a comparison between the first connection characteristics associated with the second communication and second connection characteristics associated with the first communication.
9. The storage device or storage disk of claim 3, wherein at least one of the one or more processor circuits is to initiate the sending of the second communication within a predetermined interval of the first communication.
10. A security device comprising: memory; instructions; and one or more processor circuits to execute the instructions to: prevent a first communication from reaching a receiving device, the first communication from a sending device and addressed to the receiving device; cause the security device to establish a second communication with the receiving device; collect connection characteristics associated with the second communication between the security device and the receiving device; determine a level of trustworthiness of the receiving device based on the connection characteristics associated with the second communication between the security device and the receiving device, the connection characteristics including a location of the receiving device; and after a determination that the level of trustworthiness of the receiving device meets a threshold of trustworthiness, (1) notify the sending device that a new communication addressed to the receiving device is permitted and (2) permit the new communication to reach the receiving device.
11. The security device of claim 10, wherein at least one of the one or more processor circuits is to, based on determining that the receiving device does not meet the threshold of trustworthiness, prevent the sending device from establishing the new communication with the receiving device.
12. The security device of claim 11, wherein at least one of the one or more processor circuits is to prevent the sending device from establishing the new communication with the receiving device by blocking at least one communication from the sending device to the receiving device.

13. The security device of claim 11, wherein at least one of the one or more processor circuits is to prevent the sending device from establishing the new communication with the receiving device by preventing transmission of data from the sending device to the receiving device.
14. The security device of claim 10, wherein at least one of the one or more processor circuits is to, based on determining that the receiving device does not meet the threshold of trustworthiness, instruct the sending device to not connect to the receiving device.
15. The security device of claim 10, wherein the connection characteristics are first connection characteristics, and at least one of the one or more processor circuits is to determine the level of trustworthiness of the receiving device based on the first connection characteristics associated with the second communication and second connection characteristics associated with a fourth communication including a peer device and the receiving device.
16. The security device of claim 10, wherein the connection characteristics are first connection characteristics, and at least one of the one or more processor circuits is to determine the level of trustworthiness based on a comparison between the first connection characteristics associated with the second communication and second connection characteristics associated with the first communication.
17. A method comprising: preventing, by executing an instruction with at least one processor circuit of a security device, a first communication from reaching a receiving device, the first communication from a sending device and addressed to the receiving device; establishing, by executing an instruction with one or more of the at least one processor circuit, a second communication between the security device and the receiving device; collecting, by executing an instruction with one or more of the at least one processor circuit, connection characteristics associated with the second communication between the security device and the receiving device; determining, by executing an instruction with one or more of the at least one processor circuit, a level of trustworthiness of the receiving device based on a comparison between the connection characteristics associated with the second communication and a threshold of trustworthiness; and after determining that the level of trustworthiness meets the threshold, (1) notifying by executing an instruction with one or more of the at least one processor circuit, the sending device that a third communication addressed to the receiving device is permitted and (2) permitting, by executing an instruction with one or more of the at least one processor circuit, the third communication to reach the receiving device.
18. The method of claim 17, further including, after determining that the receiving device does not meet the threshold of trustworthiness, preventing the sending device from establishing the third communication with the receiving device.
19. The method of claim 18, further including delaying the sending device from establishing the third communication with the receiving device by blocking at least one communication from the sending device to the receiving device.
20. The method of claim 18, further including delaying the sending device from establishing the third communication with the receiving device by preventing a transmission of data from the sending device to the receiving device.
21. The method of claim 17, further including, based on determining that the receiving device does not meet the threshold of trustworthiness, instructing the sending device to not connect to the receiving device.
22. The method of claim 17, wherein the connection characteristics are first connection characteristics, and the method includes determining the level of trustworthiness of the receiving device based on the first connection characteristics associated with the second communication and second connection characteristics associated with a fourth communication including a peer device and the receiving device.
23. The method of claim 17, wherein the connection characteristics are first connection characteristics, the comparison is a first comparison, and the method includes determining the level

of trustworthiness based on a second comparison between the first connection characteristics associated with the second communication and second connection characteristics associated with the first communication.

24. A security device comprising: communication circuitry; instructions; and one or more processor cores to execute the instructions to: prevent a first communication from reaching a receiving device, the first communication from a sending device and addressed to the receiving device; initiate sending of a second communication the security device to the receiving device; collect one or more characteristics associated with the second communication between the security device and the receiving device; determine a level of trustworthiness of the receiving device based on the one or more characteristics associated with the second communication between the security device and the receiving device; and after a determination that the level of trustworthiness of the receiving device meets a threshold of trustworthiness, (1) cause the communication circuitry to transmit a message to notify the sending device that a third communication addressed to the receiving device is permitted and (2) permit the third communication to reach the receiving device.

25. The security device of claim 24, wherein at least one of the one or more processor cores is to, based on determining that the receiving device does not meet the threshold of trustworthiness, prevent the sending device from establishing the third communication with the receiving device.

26. The security device of claim 24, wherein the one or more characteristics are one or more first characteristics, and at least one of the one or more processor cores is to determine the level of trustworthiness of the receiving device based on the one or more first characteristics associated with the second communication and one or more second characteristics associated with a fourth communication including a peer device and the receiving device.

27. The security device of claim 24, wherein the one or more characteristics are one or more first characteristics, and at least one of the one or more processor cores is to determine the level of trustworthiness based on a comparison between the one or more first characteristics associated with the second communication and one or more second characteristics associated with the first communication.